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SUPERSEDING
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MILITARY SPECIFICATION

INSULATION TAPE, THERMAL

This specification has been approved by the Departments of the Army, the Navy, and the Air Force

1. SCOPE

1.1 **Scope.**—This specification covers thermal insulating tape for use as an insulation for piping and tubing at temperatures up to 750° F. It is especially suitable for small piping and tubing where space conditions aboard naval vessels render awkward the use of molded insulation covering.

1.2 **Classification.** Thermal insulating tape shall be of the following types, as specified (see 6.1):

Type I—For spiral wrapping.
Type II—For lateral wrapping.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS AND PUBLICATIONS

2.1 The following specifications and standards of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

NN-B-591 —Boxes, Fiberboard Wood-Cleated, (for Domestic Shipment).
NN-B-601 —Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
NN-B-621 —Boxes, Wood, Nailed and Lock-Corner.
NN-B-631 —Boxes, Wood, Wirebound, for Domestic Shipment.
CCC-T-191 —Textile Test Methods.
LLL-B-631 —Boxes, Fiber, Corrugated (for Domestic Shipment).
LLL-B-636 —Boxes, Fiber, Solid (for Domestic Shipment).

MILITARY

JAN-P-105 —Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN-P-106 —Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
JAN-P-125 —Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.
JAN-P-139 —Packaging and Packing for Overseas Shipment — Plywood, Container-Grade.
MIL-A-140 —Adhesive, Water - Resistant, Waterproof Barrier-Material.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking of Shipments.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 **Material.** — Thermal insulating tape shall be composed of a woven asbestos jacket enclosing a woven asbestos fiber or fibrous glass felting or sliver. The fibrous glass shall be free from shot, dust, or excessive oil or moisture.

3.2 **Jacket.**

3.2.1 *Construction.* — The jacket for the hot side and not less than ¼ inch on the edges of the finished tape shall be woven from asbestos yarn containing not less than 90 percent by weight of asbestos fiber. The back or cold side of the tape shall be woven from asbetos yarn containing not less than 80 percent by weight of asbestos fiber. The dry asbestos fiber shall contain not less than 12 percent by weight of chemically combined water. The jacket may be either tubular woven in one piece or fabricated of asbestos cloth sewed into tubular form.

3.2.1.1 *Weave.*—The jacket shall be plain weave and consist of not less than 16 strands per inch in the warp and not less than 7 strands per inch in the filling. (The jacket may have an open weave on the hot side consisting of not less than 8 strands per inch in the warp and 7 strands per inch in the filling.)

3.2.2 *Breaking strength.*—The breaking strength of samples of the asbestos cloth jacket from the cold side of the tape shall be not less than 60 pounds per inch width.

3.3 **Felting or sliver.**

3.3.1 *Construction.*—The felting or sliver may be composed of one or more fabricated or felted rovings of either fibrous glass or carded amosite asbestos fibers containing not less than 95 percent by weight, of asbestos. When a single felted roving construction is used, the felting or sliver shall be suitably reinforced and held in shape as a single unit by 8 or more loosely spiralled or braided 2-ply asbestos or glass-fiber strands. When a fabrication of more than one fibrous glass or asbestos-fiber roving is used to construct the felting or sliver, the rovings shall be held in place and together by a one-over-one-under-plain weave or not less than 8 warp ends twill weave of 2-ply glass or asbestos fiber yarns spaced approximately ⅞ inch apart.

3.3.1.1 *Reinforcing strands.* — Plain and wire-inserted reinforcing strands will be allowed, if required, in the construction.

3.3.1.2 *Wire.* — The wire used in reinforcing strands shall be of brass, copper, or nickel-copper alloy, 0.008 inch in diameter, plus or minus 0.001 inch.

3.3.1.3 *Yarn.*—The asbestos or glass-fiber yarns used in the supporting structure shall be two-ply. The asbestos-fiber yarn shall contain not less than 90 percent, by weight, of asbestos fiber having not less than 12 percent, by weight, water of hydration.

3.3.1.4 *Fineness of fibers.*—The diameter of individual glass fibers shall average not more than 0.0008 inch. The maximum diameter of any fiber in the test sample shall be not more than 0.0015 inch.

3.3.2 *Alkalinity.* — The alkalinity of the fibrous glass, expressed as sodium oxide, shall not exceed 0.60 percent.

3.4 **Loss in weight.**—Tape in finished form shall not show a loss in weight greater than 15 percent, after subjection to the test specified in 4.4.7.

3.5 **Type I—for spiral wrapping.**

3.5.1 *Weave.* — The open weave of the jacket shall be not more than 90 percent of the nominal width of the material.

3.5.2 *Dimensions.* — Thermal insulating tape shall be 2 inches wide by ¼ inch thick, or 2⅜ inches wide by ⅜ inch thick, as speci-

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fied (see 6.1). A tolerance of ± 1/8 inch in width and ± 1/16 inch in thickness will be permitted.

3.5.3 *Weight.* — Thermal insulating tape shall weigh not less than 5 pounds per inch width per 100 linear feet for tape with a fibrous-glass felting or sliver, and not less than 6½ pounds per inch width per 100 linear feet for tape with an asbestos-fiber felting or sliver. The tubular jacket shall be not more than 70 percent, by weight, of the finished tape made with either a fibrous-glass or an asbestos-fiber felting or sliver.

3.6 Type II—for lateral wrapping.

3.6.1 *Weave.* — The open weave of the jacket shall be not more than 70 percent of the nominal width of the material.

3.6.2 *Dimension.* — Thermal insulating tape shall be 5¼ inches wide and 3/16 inch thick (see 6.1).

3.6.3 *Weight.* — Thermal insulating tape shall weigh not less than 25 pounds per 100 linear feet for tape with a fibrous-glass felting or sliver, and not less than 40 pounds per 100 linear feet for tape with an asbestos-fiber felting or sliver. The tubular jacket shall be not more than 55 percent by weight, of the finished tape made with either a fibrous-glass or an asbestos-fiber felting or sliver.

3.7 Marking.—The tape shall be stamped or marked in a legible manner so as to identify either the "hot side", or "cold side".

3.8 Workmanship.—The workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Inspection procedures.—For naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Lot.—For purpose of sampling, a lot shall consist of all material of one type and size offered for delivery at one time.

4.3 Inspection (at the place of manufacture).

4.3.1 *Sampling for inspection.*—A random sample of rolls shall be selected from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the sampling inspection requirements of table I in accordance with Standard MIL–STD–105.

TABLE I.—*Sampling for inspection AQL (approx.) = 2.5 percent defective.*

Number of rolls in inspection lot	Number of rolls in sample	Acceptance number (defectives)	Rejection number (defectives)
40 and under	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	2	3
501 to 800	50	3	4
801 to 1300	75	4	5
1301 to 3200	110	6	7
Over 3200	150	8	9

4.3.2 *Visual and dimensional inspection.*—Each of the sample rolls selected in accordance with 4.3.1 shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Any roll in the sample containing one or more defects shall be rejected, and if the number of defective rolls in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection provided the contractor has removed all nonconforming products. The Government inspector shall again select and examine samples from such rejected lots to verify compliance with this specification.

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4.4 Tests.

4.4.1 *Preparation of sample.* — A sample shall be formed by cutting small strips from different parts of one of the rolls selected in accordance with 4.3.1. The size of the sample shall be sufficient to provide the specimens required for the tests specified in 4.4.2 to 4.4.9, inclusive.

4.4.1.1 Unless otherwise specified herein, specimens to be tested shall be conditioned for at least 24 hours in an atmosphere having a relative humidity of 65 ± 2 percent and a temperature of 70° ± 10° F.

4.4.2 *Cotton asbestos chemically combined water.* — Separate analyses shall be performed on the warp yarns on both the hot and cold side of the jacket, the filling yarns of the jacket, yarns in the supporting structure (if not glass fiber) and the felting or sliver. Select a representative sample of yarn or felting and manually separate extraneous substances such as wire or glass fibers. Cut the residue into ⅛-inch lengths and mix thoroughly. Extract for three hours with chloroform or benzol or a Soxhlet extractor. Remove solvent and re-extract for one hour with acetone. Dry for one hour at 105° – 110° C. Weigh the absorption bulb (A). A known weight of approximately 1 gram is transferred to a porcelain or platinum boat and inserted just inside the combustion tube and stoppered firmly (see fig. 1). Maintain the combustion zone at 850° ± 25° C., the catalyst zone at 750° ± 25° C., the oxygen flow at about 200 milliliters (ml.) per minute. By means of the positioning rod of the breech mechanism, carefully move the boat forward into the combustion zone and retract rod. Continue the passage of oxygen for 15 minutes. Reweigh absorption bulb (A). The percent cotton and asbestos is determined as follows:

$$\text{Percent cotton} = \frac{61.4 \times \text{increase in weight of bulb (A)}}{\text{Weight of cleaned dried sample}}$$

Where: 61.4 = theoretical factor, 100 $\left[\frac{C_6H_{10}O_5}{6CO_2}\right]$

Percent asbestos = 100 – percent cotton

Remove the combustion boat and cool in a desiccator. Transfer the residue to balance and weigh. Chemically combined water is computed as follows:

$$\text{Percent ignited asbestos} = \frac{\text{Weight of residue} \times 100}{\text{Weight of cleaned dried sample}}$$

$$\text{Percent chemically combined water} = \frac{\text{Percent asbestos} - \text{percent ignited asbestos}}{\text{Percent asbestos}} \times 100$$

4.4.3 *Weave.*—The number of strands per inch in the warp and the filling of the jacket shall be determined as specified in Specification CCC–T–191.

4.4.4 *Dimensions.*—The width of the tape shall be determined when it is compressed to the specified nominal thickness.

4.4.5 *Weight.*—The tape shall be laid out flat. A specimen of not less than 3 linear feet shall be cut from each roll and weighed to the nearest 0.1 ounce. For type I tape, calculate the weight to pounds per inch of width per 100 linear feet using width as determined by the method specified in 4.4.4. For type II tape, calculate the weight to pounds per 100 linear feet. In both cases remove the jacket, weigh to the nearest 0.1 ounce and calculate the percent by weight of jacket in the finished tape.

4.4.6 *Breaking strength.* — The breaking strength of the asbestos cloth jacket shall be determined by the raveled-strip specified in Specification CCC–T–191 using a pendulum type machine. The jaws through which the load is applied move at a uniform rate of 12 ± 2 inches per minute. The distance between the two pairs of jaws shall be 3 inches at the start of the test. The breaking strength shall be the average result obtained

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by breaking 3 specimens of the jacket, and shall be reported as pounds per inch of width.

4.4.7 *Loss in weight.* — Compliance with 3.4 shall be determined by subjecting not less than 1 linear foot to soaking heat for 6 hours at approximately 750° F.; the weight before and after heating shall be determined.

$$\text{Loss in weight, percent} = \frac{\text{Weight before heating-weight after heating}}{\text{Weight before heating}} \times 100$$

4.4.8 *Alkalinity.*—Weigh a 5 gm. (plus or minus 0.01) representative sample of the fibrous glass, and introduce into a 500 ml. pyrex Erlenmeyer flask. Wet with 5 ml. of 95 percent ethyl alcohol, and add 400 ml. of distilled water. Reflux for 4 hours plus or minus 5 minutes. At the end of this period disconnect the condenser and filter at once using suction through No. 41 Whatman paper supported in a Buechner funnel. Wash the flask and material three times with 25 ml. portions of hot distilled water. Titrate the filtrate immediately with 0.02N H_2SO_4, using 6 to 8 drops of 1 percent solution of phenol-red indicator, to the disappearance of the pink color. Run a blank determination of the same amount of distilled water and alcohol and correct for any alkalinity shown. The percentage alkalinity as Na_2O is calculated from the following formula: percent Na_2O = 0.0124 (ml. sH_2SO_4 used by sample minus ml. sH_2SO_4 used by blank).

4.4.9 *Diameter of fibers.*—A specimen shall be taken from sample and the fiber diameter determined microscopically, based on at least five checks from each specimen. If either the maximum diameter of any one fiber thus noted, or the average of sixty such diameters exceed the corresponding limit specified in 3.3.1.4 the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Tape shall be delivered in rolls containing 25 linear feet. Each roll shall be separately wrapped with a substantial wrapping paper.

5.2 Packing.

5.2.1 *For domestic shipment.*—Tape, packaged as specified in 5.1, shall be packed in a wood-cleated-fiberboard, wood-cleated-plywood, nailed wood, wirebound wood, corrugated fiber, or solid fiber box conforming to Specification NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631, or LLL–B–636, respectively. The gross weight of wood boxes shall not exceed approximately 200 pounds. The gross weight of fiberboard boxes shall be subject to the limitation of the box specification.

5.2.2 *For oversea shipment.*—Tape, packaged as specified in 5.1, shall be overpacked in cleated plywood, or nailed wood boxes conforming to Specification JAN–P–105, style A or B (using plywood type A or B, 5 plies, condition 1 of Specification JAN–P–139), or JAN–P–106, respectively. Each box shall be lined with a sealed waterproof bag made of material conforming to Specification JAN–P–125. The seams and closures shall be made with a water resistant adhesive conforming to Specification MIL–A–140 and shall have strength and water-resistance equal to that of the body material. The gross weight of container and contents shall not exceed approximately 125 pounds.

5.3 Marking.—In addition to any special marking required by the contract or order marking shall be in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Dimensions of tape (see 3.5.2 and 3.6.2).

6.2 Thermal insulating tape should be purchased by the linear yard.

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6.3 Superseding documents.—This specification supersedes MIL–T–15349 (SHIPS). The latter superseded Navy Specification 32T1.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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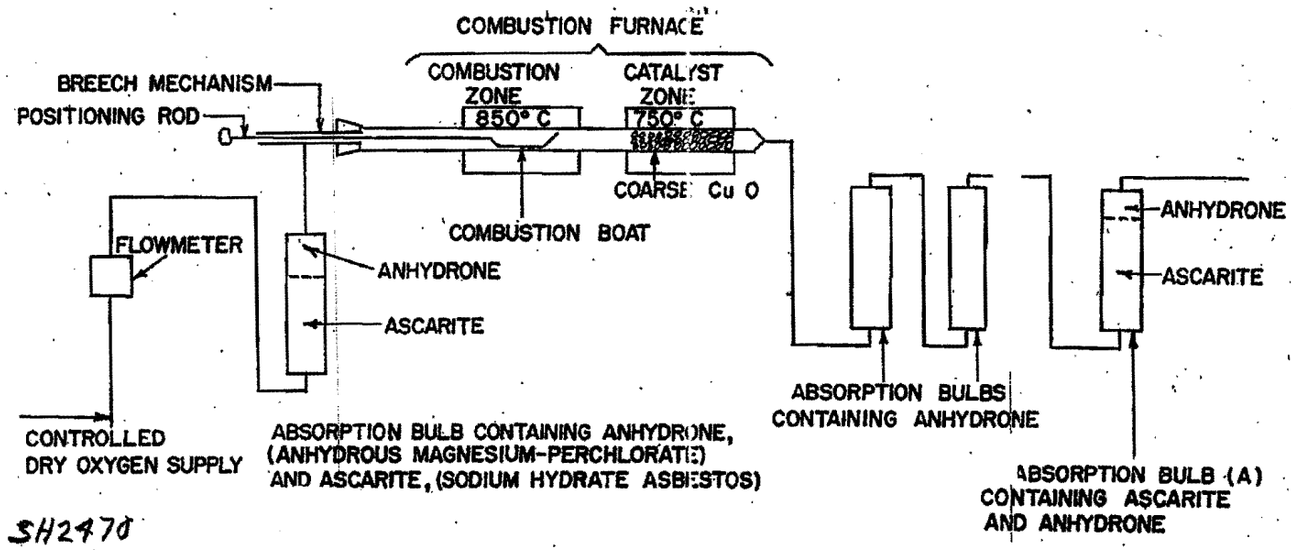


FIGURE 1.—*Apparatus for determination of cotton in asbestos-cotton samples.*

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